

Semester Paper

Deep Hedging of Path-Dependent Options

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Abstract

We study the hedging of path-dependent derivative contracts under market frictions within the deep hedging framework of [Buehler et al. \[2019\]](#). That framework recasts dynamic hedging as a stochastic optimisation problem: a neural network hedging policy is trained end-to-end to minimise a convex risk measure evaluated on simulated terminal gains, accommodating transaction costs and arbitrary payoff structures that are analytically intractable.

We implement and compare three neural network architectures as hedging policies: a position-wise multilayer perceptron (MLP), a long short-term memory network (LSTM) [[Hochreiter and Schmidhuber, 1997](#)], and a causal Transformer with multi-head self-attention [[Vaswani et al., 2017](#)] equipped with a strict lower-triangular causal mask to prevent lookahead. The risk measure is the Conditional Value-at-Risk at the 5% tail level, optimised via the differentiable dual representation of [Rockafellar and Uryasev \[2000\]](#).

Two market regimes are examined. The first is a Black-Scholes model with a floating-strike lookback call, where an analytical delta benchmark from [Goldman et al. \[1979\]](#) is available and the discrete monitoring premium is estimated by Monte Carlo to remove a continuous-monitoring bias. The second is the Heston stochastic-volatility model [[Heston, 1993](#)], simulated via the Quadratic-Exponential scheme of [Andersen \[2008\]](#), with an arithmetic Asian call and a down-and-out barrier call both hedged jointly with the stock and a variance swap.

Our central finding is that the benefit of temporal sequence models is regime-dependent. When the observable Markovian state is a sufficient statistic for the optimal hedge, all three architectures converge to similar tail-risk levels. Under Heston dynamics, where stochastic variance creates long-range temporal dependencies, the Transformer achieves a meaningful improvement in CVaR over both the MLP and the LSTM in the Asian option, whereas in the barrier option the performances of the three networks are similar.

As a further ablation, we test whether providing the policy with its own previous hedge position yields an empirical improvement in tail risk. At the transaction-cost coefficient used throughout the study, the augmentation does not improve hedging quality. In the Asian experiments it produces a substantial degradation, with the Transformer losing essentially all of its baseline advantage over the simpler architectures, while in the barrier and lookback experiments the effect is within estimation noise.

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1 Introduction

In a frictionless, complete market, arbitrage-free pricing and perfect replication are equivalent, and the Black-Scholes delta provides an exact replicating strategy for any European payoff [Black and Scholes, 1973]. In practice this solution fails on at least three counts. First, continuous rebalancing is impossible; portfolios are adjusted at discrete intervals, incurring discretisation error that grows with the length of the rebalancing interval. Second, transaction costs, proportional bid-ask spreads, market impact, and brokerage fees, make frequent rebalancing expensive, so the optimal strategy under costs changes. Third, real markets are not complete: stochastic volatility, jumps, and correlation risk create sources of uncertainty that the underlying asset alone cannot hedge.

A response to these challenges is to abandon model-driven hedging in favour of direct optimisation. Rather than deriving a strategy from a pricing formula, one directly minimises a risk measure over the space of all admissible hedging strategies. The key insight of Buehler et al. [2019] is that this optimisation problem can be solved efficiently by parametrising the hedging policy as a neural network and training it by stochastic gradient descent on simulated paths. The resulting strategy is learned from the simulated market. Crucially, this framework accommodates transaction costs, liquidity constraints, and arbitrary path-dependent payoffs without requiring any analytical tractability.

Within this *deep hedging* framework, a natural question arises that the original paper leaves largely open: how much does the choice of neural network architecture matter, and in which market regimes does temporal memory in the network provide a meaningful improvement over a memoryless baseline? The case for a Transformer, which can attend to any point in the price history via causal self-attention, over a simple position-wise MLP must rest on demonstrable tail-risk improvements.

Contributions

This paper provides a systematic answer through six controlled experiments organised in two groups. We implement three neural network architectures as hedging policies: a position-wise multilayer perceptron (MLP) as the memoryless baseline, a long short-term memory network (LSTM, Hochreiter and Schmidhuber 1997) as the standard recurrent baseline, and a causal Transformer [Vaswani et al., 2017] with scaled dot-product multi-head self-attention and a Pre-LayerNorm residual structure [Ba et al., 2016]. All three architectures share the same hidden dimension ($d = 64$) and depth (two layers), and are trained with identical hyperparameters under the Rockafellar-Uryasev CVaR objective [Rockafellar and Uryasev, 2000].

Group A: Heston experiments. Four experiments use the Heston stochastic volatility model [Heston, 1993], simulated via the Quadratic-Exponential scheme of Andersen [2008]. The agent hedges with both the stock and a variance swap [Buehler et al., 2019]. Two contracts are studied: the arithmetic Asian call and the discretely monitored down-and-out call (DOC). Each contract is run twice, once with the key path-dependent state variable (running average for Asian, running minimum for DOC) provided as an explicit feature (full setting), and once with this variable omitted (reduced setting). The four experiments test whether temporal memory in the network can recover the missing path statistic from the price history alone.

Group B: Black-Scholes experiments. Two experiments use the Black-Scholes model.

The contract is a floating-strike lookback call, discretely monitored at $n = 50$ equally spaced dates over one year. An analytical benchmark is available from the Goldman-Sosin-Gatto formula of [Goldman et al. \[1979\]](#), applied at each discrete step. The two experiments differ in whether the running minimum is provided as a feature: in the full setting, all three architectures compete against an analytical benchmark; in the reduced setting, only memory-capable architectures can recover the running minimum from the price history.

Position-aware extension. A separate ablation, reported in Chapter 7, repeats all six experiments with the policy’s own previous hedge position appended to the feature vector. The ablation tests whether the knowledge of previous positions translates into an empirical improvement in tail-risk quality.

Outline

The remainder of this paper is organised as follows. Chapter 2 develops the mathematical framework: the discrete-time market model, convex risk measures, CVaR, and the deep hedging optimisation problem. Chapter 3 specifies the three derivative contracts, the hedging instruments (stock and variance swap), and the proportional transaction cost model. Chapter 4 describes the two simulation methods: the exact log-Euler scheme for GBM and the Quadratic-Exponential scheme of [Andersen \[2008\]](#) for the Heston model, together with the parameter choices and seed convention used across all experiments. Chapter 5 presents the three neural network architectures, the feature engineering for all six experiments, and the training procedure. Chapter 6 presents the numerical results for all six experiments, organised by group. Chapter 7 reports a controlled ablation in which the policy’s own previous hedge position is appended to the feature vector across all six experiments. Chapter 8 synthesises the findings of both sets of experiments, characterises when temporal memory in the network is worth its additional complexity, and discusses why the position-aware augmentation fails to deliver on its theoretical promise at the cost level studied.

2 Mathematical Framework

We follow [Buehler et al. \[2019\]](#), Sections 2 and 3, restricting attention to what the numerical experiments require.

2.1 The Financial Market

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a finite probability space, fix a finite horizon $T > 0$ and trading dates $0 = t_0 < t_1 < \dots < t_n = T$, and let $\mathbb{F} = (\mathcal{F}_k)_{k=0, \dots, n}$ with $\mathcal{F}_n = \mathcal{F}$ model the progressive revelation of information.

Assumption 2.1. At each date t_k the agent observes an information set $I_k \in \mathbb{R}^r$ that generates $\mathcal{F}_k$; in addition to prices, I_k may include implied volatility, accrued path statistics, or any other $\mathcal{F}_k$ -measurable quantity.

The market contains $d \geq 1$ hedging instruments with mid-prices given by an $\mathbb{R}^d$ -valued $\mathbb{F}$ -adapted process $S = (S_k)$. Following [Buehler et al. \[2019\]](#), no equivalent martingale measure is assumed: with transaction costs and market incompleteness, perfect replication is generically impossible and no derivative pricing model is required to compute hedge ratios. The risk-free rate is set to zero throughout, a standard simplification in the deep hedging literature.

Remark 2.2 (Simulation under the risk-neutral measure). The numerical experiments are conducted under the risk-neutral measure $\mathbb{Q}$. Two practical reasons: it permits comparison with the analytical benchmarks (the Goldman-Sosin-Gatto delta for the lookback, the no-arbitrage premium for the Asian); and the variance swap used as the second hedging instrument in the Heston experiments is priced as

$$\text{VS}_t = \int_0^t V_s ds + \mathbb{E}^{\mathbb{Q}} \left[\int_t^T V_s ds \mid V_t \right], \quad (2.1)$$

following [Buehler et al. \[2019\]](#), equations (5.3)–(5.4)], which uses the risk-neutral CIR parameters. Setting $r = 0$ makes S a $\mathbb{Q}$ -martingale; each Monte Carlo premium is computed as $p_0 = \mathbb{E}^{\mathbb{Q}}[\text{payoff}]$ on a dedicated pool of pricing paths.

2.2 Trading Strategies and the Terminal Gain

Definition 2.3 (Trading strategy, [Buehler et al. 2019](#) Section 2). A trading strategy is an $\mathbb{R}^d$ -valued $\mathbb{F}$ -adapted process $\delta = (\delta_k)_{k=0, \dots, n-1}$, with δ_k^i the number of units of instrument i held between t_k and t_{k+1} . By convention $\delta_{-1} = \delta_n = 0$: no position before the first trading date, fully liquidated at maturity. The set of unconstrained such strategies is $\mathcal{H}^u$.

The cumulative trading P&L of δ is

$$(\delta \cdot S)_n := \sum_{k=0}^{n-1} \delta_k \cdot (S_{k+1} - S_k). \quad (2.2)$$

Transaction costs. Rebalancing at t_k costs

$$c_k(\delta_k - \delta_{k-1}) := c \sum_{i=1}^d S_k^i |\delta_k^i - \delta_{k-1}^i|, \quad (2.3)$$

with $c > 0$ proportional to the half-spread. The total cost is

$$C_n(\delta) := \sum_{k=0}^{n-1} c_k(\delta_k - \delta_{k-1}). \quad (2.4)$$

The terminal gain. The agent sells a derivative with terminal payoff $Z_T \geq 0$ and collects a premium $p_0 \geq 0$ at time zero. The terminal gain of the hedged position is

$$G_T(\delta) := p_0 + (\delta \cdot S)_n - C_n(\delta) - Z_T. \quad (2.5)$$

Perfect replication corresponds to $G_T \equiv 0$, generically unattainable under frictions.

2.3 Conditional Value-at-Risk

Definition 2.4 (Conditional Value-at-Risk). *For $\alpha \in (0, 1)$ and a random gain $X \in \mathcal{X}$, the Value-at-Risk is the negative lower α -quantile, $\text{VaR}_\alpha(X) := -\inf\{x : \mathbb{P}(X \leq x) \geq \alpha\}$, and the CVaR is*

$$\text{CVaR}_\alpha(X) := -\mathbb{E}[X \mid X \leq -\text{VaR}_\alpha(X)], \quad (2.6)$$

the expected loss in the worst α -fraction of scenarios.

CVaR is a coherent risk measure and is strictly more tail-sensitive than VaR. The key property for stochastic gradient descent is its dual representation.

Theorem 2.5 (Rockafellar-Uryasev, [Rockafellar and Uryasev 2000](#) Theorem 1). *For any $X \in \mathcal{X}$ and $\alpha \in (0, 1)$,*

$$\text{CVaR}_\alpha(X) = \min_{\eta \in \mathbb{R}} \left\{ \eta + \frac{1}{\alpha} \mathbb{E}[(-X - \eta)^+] \right\}, \quad (2.7)$$

the minimum attained at $\eta^ = \text{VaR}_\alpha(X)$.*

The objective $(\eta, \delta) \mapsto \eta + \alpha^{-1} \mathbb{E}[(-G_T(\delta) - \eta)^+]$ is jointly convex (for linear policies) and differentiable almost everywhere, so η can be optimised jointly with the network weights and the expectation can be approximated by a Monte Carlo average. All experiments use $\alpha = 0.05$.

2.4 The Deep Hedging Problem

Definition 2.6 (Deep hedging problem, [Buehler et al. 2019](#) equation (3.1)). *Given a liability Z_T , a premium p_0 , and a convex risk measure ρ , the optimal hedging value is*

$$\pi(Z_T) := \inf_{\delta \in \mathcal{H}^u} \rho(-G_T(\delta)). \quad (2.8)$$

An admissible strategy δ^ achieving the infimum is called an optimal hedging strategy.*

Neural network approximation. The infimum in (2.8) is over the infinite-dimensional space $\mathcal{H}^u$. [Buehler et al. \[2019\]](#) approximate it by a parametric policy network $\delta_\theta : I_k \mapsto \delta_k$ with trainable parameters θ . With $\rho = \text{CVaR}_\alpha$, Theorem 2.5 gives

$$\min_{\theta, \eta \in \mathbb{R}} \left\{ \eta + \frac{1}{\alpha} \mathbb{E}[(-G_T(\delta_\theta) - \eta)^+] \right\}, \quad (2.9)$$

with the expectation approximated by a Monte Carlo average over simulated paths and η as a scalar parameter optimised jointly with θ . The AdamW optimiser [[Loshchilov and Hutter, 2019](#)] is used for gradient updates, with mini-batches drawn from a pre-simulated path pool at each epoch.

3 Financial Framework

The six experiments use three path-dependent contracts (lookback, Asian, down-and-out) hedged with the stock alone in the Black-Scholes experiments and jointly with the stock and a variance swap in the Heston experiments, all under proportional transaction costs.

3.1 Dynamic Delta Hedging

In a frictionless complete market, a European payoff $Z_T = f(S_T)$ is replicable by the Black-Scholes delta $\Delta_t = \partial C / \partial S_t$ [Shreve, 2004, Theorem 5.4.2]. Path-dependent payoffs $Z_T = f(S_{t_0}, \dots, S_{t_n})$ break this: the replicating delta at t_k depends on an accumulated path statistic. Static positions cannot track such products; dynamic rebalancing is mandatory. The deep hedging framework parametrises the sequence $(\delta_{t_0}, \dots, \delta_{t_{n-1}})$ as a neural network and minimises the CVaR of the terminal loss as in Definition 2.6 of Chapter 2.

3.2 Derivative Contracts

Each contract is discretely monitored at n equally spaced dates; the specific T and n for each experiment appear in Chapter 6.

3.2.1 Floating-Strike Lookback Call

A floating-strike lookback call grants the holder the right to purchase the underlying at the lowest price observed at any monitoring date. The terminal payoff under discrete monitoring is

$$Z_T^{\text{LB}} = S_{t_n} - \min_{0 \leq k \leq n} S_{t_k}. \quad (3.1)$$

Since the floating strike is the realised minimum, $Z_T^{\text{LB}} \geq 0$ almost surely: the option is always at the money or in the money. The key path-dependent state variable at step k is the running minimum

$$m_{t_k} = \min_{0 \leq j \leq k} S_{t_j}. \quad (3.2)$$

Under continuous monitoring [Goldman et al., 1979], the pair (S_{t_k}, m_{t_k}) is a sufficient statistic for both price and delta; the discrete analogue inherits this approximately, so the natural feature vector is $(S_{t_k}, m_{t_k}, \tau_k)$ with $\tau_k = (n - k)/n$ used in Experiments 5 and 6 (Group B). An analytical benchmark is available from the Goldman-Sosin-Gatto delta of Goldman et al. [1979], computed via central finite differences.

3.2.2 Arithmetic Asian Call

An arithmetic Asian call has a payoff based on the arithmetic average of the underlying price over the monitoring dates. The terminal payoff is

$$Z_T^{\text{AS}} = \left(\frac{1}{n} \sum_{k=1}^n S_{t_k} - K \right)^+, \quad (3.3)$$

where K is the fixed strike price and $(x)^+ = \max(x, 0)$. The average is computed over the n post-initial dates $t_1, \dots, t_n$, excluding t_0 . The running average at step k is

$$A_{t_k} = \frac{1}{k} \sum_{j=1}^k S_{t_j}, \quad k = 1, 2, \dots, n. \quad (3.4)$$

Providing A_{t_k} as a feature (full set) versus omitting it (reduced set) is the only structural difference between Experiments 1 and 2 (Group A, Heston). No closed-form delta is available under stochastic volatility, so the neural networks are evaluated against each other.

3.2.3 Down-and-Out Call

The down-and-out call (DOC) behaves as a European call unless the underlying touches a lower barrier $H < K$ at any monitoring date, in which case it knocks out:

$$Z_T^{\text{DOC}} = (S_{t_n} - K)^+ \cdot \mathbf{1} \left\{ \min_{0 \leq k \leq n} S_{t_k} > H \right\}. \quad (3.5)$$

The state variable is again m_{t_k} : once $m_{t_k} \leq H$ all future hedging is irrelevant. In the experiments, $K = 1.0$ and $H = 0.85$. Used in Experiments 3 and 4 (Group A, Heston). A closed-form price is available under constant volatility and continuous monitoring [Merton, 1973], but no closed-form delta exists for the discretely monitored version under Heston dynamics.

Remark 3.1 (Discrete versus continuous monitoring). All premiums are Monte Carlo estimates of the *discrete* payoff expectation. Using a continuous-monitoring premium for the upfront cash flow against a discrete payoff at maturity would shift the gain distribution by the monitoring gap, breaking the consistency of the gain formula.

3.3 Hedging Instruments

3.3.1 The Underlying Stock

In all experiments the stock is a hedging instrument. The agent holds $\delta_{t_k}^S$ units between t_k and t_{k+1} ; the cumulative trading gain is $(\delta \cdot S)_n$ from equation (2.2).

3.3.2 The Variance Swap

In the Heston experiments the agent also hedges with a variance swap, an OTC contract whose terminal payoff is the realised variance of the underlying,

$$\text{Payoff at } T = \int_0^T V_s ds, \quad (3.6)$$

where V_s is the instantaneous variance of the stock [Buehler et al., 2019, equations (5.3)–(5.4)]. Two reasons make it the natural second instrument: V_t is not directly observable from the stock price, and the variance swap is a tradeable proxy for it; and the path-dependent payoffs are volatility-sensitive, so the stock alone cannot hedge their vega exposure.

The fair value at t given accumulated realised variance and current V_t is

$$\text{VS}_t = \int_0^t V_s ds + L(t, V_t), \quad (3.7)$$

which expands (2.1) with

$$L(t, v) = \theta(T - t) + (v - \theta) \frac{1 - e^{-\kappa(T-t)}}{\kappa}, \quad (3.8)$$

following [Buehler et al. \[2019\]](#), equations (5.3)–(5.4)], where θ and κ are the long-run mean and mean-reversion speed of the CIR variance process. At maturity $L(T, v) = 0$ and VS_T matches (3.6).

In the Heston experiments the agent chooses two-dimensional positions $(\delta_{t_k}^S, \delta_{t_k}^{\text{VS}})$ and the gain formula (2.5) applies coordinate-wise.

3.4 Market Frictions

Transaction costs are the primary friction. Following [Buehler et al. \[2019\]](#), equation (2.1)], every rebalancing at t_k incurs

$$c_k(\delta_k - \delta_{k-1}) = c S_{t_k} |\delta_k^S - \delta_{k-1}^S| \quad (3.9)$$

in the single-instrument case, and

$$c_k(\delta_k - \delta_{k-1}) = c S_{t_k} |\delta_k^S - \delta_{k-1}^S| + c |\text{VS}_{t_k}| |\delta_k^{\text{VS}} - \delta_{k-1}^{\text{VS}}| \quad (3.10)$$

in the two-instrument Heston case, with the variance-swap leg scaled by $|\text{VS}_{t_k}|$ since VS_t is in variance units.

Transaction costs break perfect replication: $C_n(\delta)$ is path-dependent and non-linear in δ , so no closed-form minimiser exists. They also create an explicit trade-off between hedging accuracy and trading frequency, which the CVaR objective resolves directly by optimising over the empirical loss distribution.

All six experiments use $c = 0.001$, corresponding to a half-spread of ten basis points, consistent with the liquid equity index market assumed by [Buehler et al. \[2019\]](#). At this cost level the no-trade region around the frictionless delta is narrow; whether widening the policy's information set with its own previous hedge position improves tail-risk quality is the subject of Chapter 7.

4 Path Simulation

The Black-Scholes experiments use geometric Brownian motion (GBM); the Heston experiments use the Heston stochastic volatility model, simulated by the Quadratic-Exponential (QE) scheme of Andersen [2008], which guarantees non-negative variance and matches the first two conditional moments of the continuous-time distribution.

4.1 Geometric Brownian Motion

Under the risk-neutral measure $\mathbb{Q}$ with $r = 0$, the stock price follows

$$\frac{dS_t}{S_t} = \sigma dW_t, \quad (4.1)$$

with $\sigma > 0$ and W a standard $\mathbb{Q}$ -Brownian motion. The exact log-Euler step

$$S_{t_{k+1}} = S_{t_k} \exp\left[-\frac{1}{2}\sigma^2\Delta t + \sigma\sqrt{\Delta t} Z_{k+1}\right], \quad Z_{k+1} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1), \quad (4.2)$$

reproduces the joint law of $(S_{t_0}, \dots, S_{t_n})$ without discretisation error [Glasserman, 2003, Section 3.2]. Paths are stored in a matrix $\mathbf{S} \in \mathbb{R}^{N \times (n+1)}$.

Table 1: Simulation parameters for the Black-Scholes experiments (Group B, Experiments 5 and 6).

Parameter	Symbol	Value
Initial price	S_0	1.0
Strike price	K	1.0 (ATM)
Risk-free rate	r	0.0
Volatility	σ	0.20
Maturity	T	1.0 year
Hedging steps	n	50 (weekly)
Step size	Δt	0.02 years

Setting $r = 0$ makes S a $\mathbb{Q}$ -martingale and gives the cleaner $r \rightarrow 0$ limit of the Goldman-Sosin-Gatto closed-form delta. $\sigma = 0.20$ is the standard equity-market calibration of Buehler et al. [2019].

4.2 The Heston Stochastic Volatility Model

The Heston model [Heston, 1993] adds a stochastic variance process to GBM. Under $\mathbb{Q}$ the joint dynamics of the log-price and variance are

$$d \ln S_t = -\frac{1}{2} V_t dt + \sqrt{V_t} dW_t^S, \quad (4.3)$$

$$dV_t = \kappa(\theta - V_t) dt + \varepsilon \sqrt{V_t} dW_t^V, \quad (4.4)$$

with mean-reversion speed $\kappa > 0$, long-run mean variance $\theta > 0$, vol-of-vol $\varepsilon > 0$, and correlation $dW_t^S dW_t^V = \rho dt$. The negative ρ captures the leverage effect.

The variance is a CIR process [Cox et al., 1985], strictly positive under the Feller condition $2\kappa\theta \geq \varepsilon^2$. For the DOC parameters of Table 3 this gives $2\kappa\theta = 0.16$ against $\varepsilon^2 = 0.25$, so the condition fails: the variance can reach zero, which makes the discretisation choice critical.

Table 2: Simulation parameters for the Asian call experiments (Group A, Experiments 1 and 2).

Parameter	Symbol	Value
Initial price	S_0	100.0
Strike price	K	100.0 (ATM)
Risk-free rate	r	0.0
Initial variance	V_0	0.04 (= 20% vol)
Mean-reversion speed	κ	1.0
Long-run variance	θ	0.04 (= 20% vol)
Volatility of vol	ε	2.0
Spot-vol correlation	ρ	-0.70
Maturity	T	30/365 years
Hedging steps	n	30 (daily)
Step size	Δt	1/365 years

Table 3: Simulation parameters for the down-and-out call experiments (Group A, Experiments 3 and 4).

Parameter	Symbol	Value
Initial price	S_0	1.0
Strike price	K	1.0 (ATM)
Barrier level (DOC)	H	0.85
Risk-free rate	r	0.0
Initial variance	V_0	0.04 (= 20% vol)
Mean-reversion speed	κ	2.0
Long-run variance	θ	0.04 (= 20% vol)
Volatility of vol	ε	0.50
Spot-vol correlation	ρ	-0.70
Maturity	T	1.0 year
Hedging steps	n	50 (weekly)
Step size	Δt	0.02 years

4.3 The Quadratic-Exponential Scheme

The standard Euler-Maruyama scheme can yield negative $\hat{V}$ when the variance approaches zero, requiring an ad-hoc reflection or absorption fix. The QE scheme [Andersen, 2008] instead samples $V_{t_{k+1}}$ from an approximation of its true conditional distribution given V_{t_k} , branching on the variance-to-mean-squared ratio $\psi = \text{Var}[V_{t_{k+1}} | V_{t_k}] / (\mathbb{E}[V_{t_{k+1}} | V_{t_k}])^2$.

4.3.1 Conditional Moments of the CIR Process

The exact conditional moments are [Andersen, 2008, equations (1.3)–(1.4)]

$$m = \mathbb{E}[V_{t+\Delta t} | V_t = v] = \theta + (v - \theta) e^{-\kappa \Delta t}, \quad (4.5)$$

$$s^2 = \text{Var}[V_{t+\Delta t} | V_t = v] = v c_1 + c_2, \quad (4.6)$$

with

$$c_1 = \frac{\varepsilon^2 e^{-\kappa \Delta t} (1 - e^{-\kappa \Delta t})}{\kappa}, \quad c_2 = \frac{\theta \varepsilon^2 (1 - e^{-\kappa \Delta t})^2}{2\kappa}. \quad (4.7)$$

The branch is chosen path-wise based on $\psi = s^2/m^2$.

4.3.2 Low-Psi Branch: Quadratic Approximation

When $\psi \leq \psi_c = 1.5$ the conditional distribution is unimodal and approximately symmetric. The QE scheme uses a scaled shifted chi-squared,

$$V^{\text{quad}} = a(\sqrt{b^2} + Z_v)^2, \quad Z_v \sim \mathcal{N}(0, 1), \quad (4.8)$$

with

$$b^2 = \frac{2}{\psi} - 1 + \sqrt{\frac{2}{\psi} \left(\frac{2}{\psi} - 1 \right)}, \quad a = \frac{m}{1 + b^2}, \quad (4.9)$$

which matches m and s^2 exactly.

4.3.3 High-Psi Branch: Exponential Approximation

When $\psi > \psi_c$ the variance spends meaningful time near zero. The scheme uses a point-mass-plus-exponential mixture,

$$V^{\text{exp}} = \begin{cases} 0 & \text{with probability } p, \\ \frac{1}{\beta} \ln \frac{1-p}{1-U} & \text{with probability } 1-p, \end{cases} \quad (4.10)$$

with

$$p = \frac{\psi - 1}{\psi + 1}, \quad \beta = \frac{2}{m(\psi + 1)}, \quad (4.11)$$

again matching the first two moments.

4.3.4 Log-Price Step

Given V_{t_k} and the sampled $V_{t_{k+1}}$, the log-price increment is [Andersen, 2008, equations (2.13)–(2.17)]

$$\ln S_{t_{k+1}} = \ln S_{t_k} + r\Delta t + K_0 + K_1 V_{t_k} + K_2 V_{t_{k+1}} + \sqrt{K_3 V_{t_k} + K_4 V_{t_{k+1}}} Z_s, \quad (4.12)$$

with $Z_s \sim \mathcal{N}(0, 1)$ independent of Z_v and U , and constants

$$\begin{aligned} K_0 &= -\frac{\rho\kappa\theta}{\varepsilon}\Delta t, \\ K_1 &= \gamma_1 \left(\frac{\rho\kappa}{\varepsilon} - \frac{1}{2} \right) \Delta t - \frac{\rho}{\varepsilon}, \quad K_2 = \gamma_2 \left(\frac{\rho\kappa}{\varepsilon} - \frac{1}{2} \right) \Delta t + \frac{\rho}{\varepsilon}, \\ K_3 &= \gamma_1(1 - \rho^2)\Delta t, \quad K_4 = \gamma_2(1 - \rho^2)\Delta t, \end{aligned}$$

which incorporate the spot/variance correlation through a trapezoidal approximation of integrated variance, $\int_{t_k}^{t_{k+1}} V_s ds \approx \Delta t(\gamma_1 V_{t_k} + \gamma_2 V_{t_{k+1}})$, with $\gamma_1 = \gamma_2 = \frac{1}{2}$.

4.4 Variance Swap Price Path

The variance swap fair value at t_k is computed from the simulated path using equation (3.7). The accumulated realised variance is updated step by step with the same trapezoidal approximation as the log-price step:

$$\int_0^{\widehat{t_{k+1}}} V_s ds = \int_0^{\widehat{t_k}} V_s ds + \Delta t(\gamma_1 V_{t_k} + \gamma_2 V_{t_{k+1}}), \quad (4.13)$$

with $\gamma_1 = \gamma_2 = \frac{1}{2}$. The forward-looking term $L(t_k, V_{t_k})$ from (3.8), $\mathbb{E}^{\mathbb{Q}}[\int_{t_k}^T V_s ds \mid V_{t_k}]$, is computed in a single vectorised operation across all paths and steps.

4.5 Simulation Parameters and Seed Convention

4.5.1 Path Pools

Three independent path pools per experiment, each with a fixed seed for full reproducibility:

- (1) *Pricing pool* (seed = 0): $N_{\text{price}} = 200,000$ paths for the Asian, 100,000 for the barrier and lookback. Used only to compute the Monte Carlo premium p_0 .
- (2) *Training pool* (seed = 1): $N_{\text{train}} = 50,000$ paths. Mini-batches of 10,000 are drawn uniformly at random at each epoch; the same pool is reused throughout training.
- (3) *Evaluation pool* (seed = 2): $N_{\text{eval}} = 10,000$ held-out paths, shared across all three architectures so that CVaR differences reflect the policy and not sampling.
- (4) *Test pool* (seed = 3): $N_{\text{test}} = 50,000$ held-out paths, never used during training or checkpoint selection. All final reported numbers in Chapter 6 come from this pool after best-checkpoint restoration.

Table 4: Path pool sizes and seeds used in all six experiments.

Pool	Seed	Paths	Purpose
Pricing	0	200,000 (Asian) / 100,000 (others)	Monte Carlo premium
Training	1	50,000	Gradient updates
Evaluation	2	10,000	Checkpoint selection during training
Test	3	50,000	Final reported results

4.5.2 Feature Matrices

Feature matrices for all three architectures are built once from the raw price (and variance swap) tensors and shared across models within an experiment, so no representational inconsistency can confound the comparison. The feature vectors are listed in Section 5.5. The same path pools and feature matrices are reused without modification in the ablation reported in Chapter 7; that chapter differs from the experiments of Chapter 6 only in appending the policy’s own previous hedge position to each feature vector.

5 Neural Network Architectures

The three architectures differ along a single axis: how much of the history $\mathcal{I}_{t_k}$ is accessible when the hedge ratio at step k is produced. Hidden dimension (64), depth (two layers), training objective, optimiser, hyperparameters, and feature matrices are held constant throughout, so any performance difference reflects the architecture’s inductive bias rather than capacity or training advantages.

5.1 The Hedging Policy and Training Objective

5.1.1 Policy Parametrisation

The hedging policy at step k is a measurable map $\delta_\theta : \mathcal{I}_{t_k} \mapsto \mathbb{R}^d$, with $d = 1$ for the lookback experiments and $d = 2$ in the two-instrument Heston experiments (stock and variance swap). Causality requires

$$\delta_{t_k}^\theta = \delta_\theta(\mathcal{I}_{t_k}), \quad \mathcal{I}_{t_k} = (\mathbf{x}_{t_0}, \dots, \mathbf{x}_{t_k}), \quad (5.1)$$

where $\mathbf{x}_{t_j} \in \mathbb{R}^d$ is the feature vector at step j (Section 5.5). The MLP uses only $\mathbf{x}_{t_k}$; the LSTM compresses the history into a hidden state; the Transformer attends to any subset of past observations.

The terminal gain $G_T(\delta)$ and the proportional transaction cost coefficient $c = 0.001$ are defined as in Chapter 2; the premium p_0 is the Monte Carlo estimate of Section 4.5.

5.1.2 CVaR Objective and the Rockafellar-Uryasev Dual Form

The training objective is $\text{CVaR}_\alpha(L_T)$ at $\alpha = 0.05$, with $L_T = -G_T(\delta)$. By Theorem 2.5 the dual representation is

$$\text{CVaR}_\alpha(L_T) = \min_{\eta \in \mathbb{R}} \left\{ \eta + \frac{1}{\alpha} \mathbb{E}[\max(L_T - \eta, 0)] \right\}, \quad (5.2)$$

which is convex in both η and the policy parameters, so joint stochastic gradient descent applies. The scalar η is a learnable parameter initialised at zero. The empirical objective is

$$\widehat{\text{CVaR}}_\alpha(\delta_\theta; \eta) = \eta + \frac{1}{\alpha N_{\text{batch}}} \sum_{i=1}^{N_{\text{batch}}} \max(-G_T^{(i)}(\delta_\theta) - \eta, 0), \quad (5.3)$$

evaluated on a uniformly sampled mini-batch of $N_{\text{batch}} = 10,000$ paths.

5.2 Position-wise MLP (Memoryless Baseline)

Two hidden layers of width $h = 64$ with Tanh activations and a linear output head, applied position-wise: at step k the network sees only $\mathbf{x}_{t_k}$ and produces δ_{t_k} . Weight sharing across steps makes the policy stationary in time. The hidden dimension matches that of the LSTM and Transformer so capacity is not the confound; parameter counts appear in Table 6.

5.3 Long Short-Term Memory Network

The LSTM [Hochreiter and Schmidhuber, 1997] processes the feature sequence step by step, maintaining a hidden state and a cell state of dimension 64 each that compress the

history into a fixed-size representation. Standard input, forget, cell, and output gates control which past information is retained, discarded, or exposed at every step.

Two layers are stacked; a linear output head maps the second-layer hidden state to the hedge ratio. Causality holds by construction.

Parameter count. With d input features and hidden size $h = 64$, the two-layer LSTM has $256d + 50,241$ parameters in the single-instrument variant (Table 6).

5.4 Causal Transformer

The Transformer [Vaswani et al., 2017] processes the entire feature sequence in one parallel forward pass and enforces causality through a lower-triangular attention mask. The Pre-LayerNorm convention [Ba et al., 2016] is used for training stability.

5.4.1 Input Projection and Positional Embeddings

The raw feature $\mathbf{x}_{t_k}$ is projected into the model dimension $d_{\text{model}} = 64$,

$$\tilde{\mathbf{x}}_k = W_{\text{proj}} \mathbf{x}_{t_k} + \mathbf{b}_{\text{proj}}, \quad W_{\text{proj}} \in \mathbb{R}^{64 \times d}, \quad (5.4)$$

and a learned positional embedding $\mathbf{e}_k \in \mathbb{R}^{64}$, indexed by step k , is added to encode temporal order:

$$\mathbf{z}_k^{(0)} = \tilde{\mathbf{x}}_k + \mathbf{e}_k. \quad (5.5)$$

Learned embeddings are preferred over the sinusoidal schedule of Vaswani et al. [2017] because the horizon is short and fixed ($n \leq 50$).

5.4.2 Causal Multi-Head Self-Attention

Each block computes scaled dot-product multi-head self-attention [Vaswani et al., 2017, Section 3.2]. Queries, keys, and values are linear projections of the sequence,

$$Q = \mathbf{Z} W_Q^\top, \quad K = \mathbf{Z} W_K^\top, \quad V = \mathbf{Z} W_V^\top, \quad (5.6)$$

with W_Q, W_K bias-free (a constant additive shift to attention scores leaves the softmax invariant). The causal mask M sets all future-position scores to $-\infty$ before the softmax,

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + M\right)V, \quad (5.7)$$

where $d_k = 64/n_{\text{heads}} = 16$ and $n_{\text{heads}} = 4$. Multi-head splitting, application of (5.7) per head, concatenation, and a final output projection W_O complete the sublayer. Output position j depends only on positions $0, \dots, j$.

5.4.3 Pre-LN Residual Blocks

Each of the $L = 2$ blocks applies attention and a position-wise feed-forward network as residual sublayers with layer normalisation inside the residual branch:

$$\mathbf{Z}' = \mathbf{Z} + \text{MHSA}(\text{LN}(\mathbf{Z})), \quad (5.8)$$

$$\mathbf{Z}'' = \mathbf{Z}' + \text{FFN}(\text{LN}(\mathbf{Z}')). \quad (5.9)$$

The feed-forward sublayer is $\text{FFN}(\mathbf{z}) = W_2 \text{GELU}(W_1 \mathbf{z} + \mathbf{b}_1) + \mathbf{b}_2$ with expansion dimension $d_{\text{ff}} = 256 = 4 d_{\text{model}}$ following the original ratio of Vaswani et al. [2017], and a smooth GELU activation [Hendrycks and Gimpel, 2016].

5.4.4 Output Head

After the $L = 2$ blocks a final layer normalisation is applied (required by Pre-LN), and a linear output head produces the hedge ratio:

$$\delta_{t_k}^{\text{Transformer}} = \mathbf{w}_{\text{out}}^\top \text{LN}(\mathbf{z}_k^{(L)}) + b_{\text{out}}. \quad (5.10)$$

No activation: hedge ratios are unconstrained.

Remark 5.1 (Parameter counts and model comparison). The three models are matched by hidden dimension and depth, not by parameter count: the LSTM and Transformer carry the overhead of their gating mechanism and feed-forward expansion.

5.5 Feature Engineering

At each step t_k the observable state is encoded in a d -dimensional feature vector $\mathbf{x}_{t_k}$, $d \in \{2, \dots, 5\}$; the experiment-by-experiment breakdown is in Table 5.

Table 5: Feature vectors for each experiment. S_{t_k} : current stock price; VS_{t_k} : variance swap fair value; m_{t_k} : running minimum; A_{t_k} : running average; H : barrier; K : strike; S_0 : initial price; VS_0 : initial variance swap value; $\tau_k = (n - k)/n$: normalised remaining time.

Exp.	Contract	Model	Features	d
1	Asian call (full)	Heston	S_{t_k}/S_0 , $\text{VS}_{t_k}/\text{VS}_0$, A_{t_k}/S_0 , τ_k	4
2	Asian call (reduced)	Heston	S_{t_k}/S_0 , $\text{VS}_{t_k}/\text{VS}_0$, τ_k	3
3	DOC barrier (full)	Heston	S_{t_k}/S_0 , $\text{VS}_{t_k}/\text{VS}_0$, m_{t_k}/H , S_{t_k}/H , τ_k	5
4	DOC barrier (reduced)	Heston	S_{t_k}/S_0 , $\text{VS}_{t_k}/\text{VS}_0$, S_{t_k}/H , τ_k	4
5	Lookback (full)	GBM	S_{t_k}/K , m_{t_k}/K , τ_k	3
6	Lookback (reduced)	GBM	S_{t_k}/K , τ_k	2

Each feature is normalised to keep magnitudes comparable across steps and paths. The variance-swap feature is normalised by its initial value $\text{VS}_0 = L(0, V_0)$ from equation (3.8); the running minimum is normalised by H (barrier) or K (lookback); $\tau_k = (n - k)/n$ is dimensionless.

The *full* feature sets include the primary path-dependent state variable (A_{t_k} for the Asian, m_{t_k} for the barrier and lookback); the *reduced* sets omit it. This ablation tests whether the LSTM and Transformer can recover the omitted statistic from the observed price history, a question the MLP cannot answer.

Feature matrices are built once from the raw path tensors and shared across all three architectures in each experiment.

An ablation in which the previous hedge position is appended to every feature vector across all six experiments, motivated by the sufficient-statistic argument under proportional transaction costs [Buehler et al., 2019], is reported in Chapter 7. That chapter reuses the architectures, hidden dimensions, and training hyperparameters of the present chapter without modification; only the input feature width changes.

5.6 Training Setup

All models train for 3,000 epochs on $N_{\text{train}} = 50,000$ paths (seed 1) using AdamW [Loshchilov and Hutter, 2019] with learning rate 10^{-3} and weight decay 10^{-4} . The scalar

Table 6: Parameter counts by architecture and feature dimension d . Single-instrument variants; two-instrument variants add one output unit. Transformer counts assume $n_{\max} = 51$.

Architecture	$d = 2$ (Exp. 6)	$d = 3$ (Exp. 2, 5)	$d = 4$ (Exp. 1, 4)	$d = 5$ (Exp. 3)
MLP	4,417	4,481	4,545	4,609
LSTM	50,753	51,009	51,265	51,521
Transformer	103,361	103,425	103,489	103,553

η from (5.2) is placed in a separate parameter group with weight decay set to zero, since penalising the VaR threshold would bias the CVaR estimator. At each epoch a mini-batch of $N_{\text{batch}} = 10,000$ paths is drawn uniformly from the training pool and a single gradient step is taken on (5.3). No learning rate schedule, gradient clipping, or dropout is used.

Every 200 epochs the policy is evaluated on a held-out pool of $N_{\text{eval}} = 10,000$ paths (seed 2); the model state at the epoch with the lowest eval CVaR is restored before the final evaluation on the test pool (seed 3, $N_{\text{test}} = 50,000$). This checkpoint mechanism prevents reported results from being contaminated by late-epoch instabilities.

Table 7: Shared training hyperparameters for all six experiments.

Hyperparameter	Value
Optimiser	AdamW [Loshchilov and Hutter, 2019]
Learning rate	10^{-3}
Weight decay (network)	10^{-4} (decoupled)
Weight decay (η)	0
Mini-batch size	10,000 paths
Training epochs	3,000
CVaR tail probability	$\alpha = 0.05$
Transaction cost	$c = 0.001$ (proportional)
Training paths	$N_{\text{train}} = 50,000$ (seed 1)
Evaluation paths	$N_{\text{eval}} = 10,000$ (seed 2)

Analytical benchmark. The Goldman-Sosin-Gatto delta [Goldman et al., 1979], evaluated at each discrete step as in Chapter 3, serves as a reference for the two lookback experiments. It is evaluated on the same test paths as the neural network policies, without training. No analytical benchmark exists for the Heston experiments under stochastic volatility, so the Asian and barrier comparisons are network against network only.

6 Numerical Experiments

Six experiments are reported, organised into two groups by underlying model. Group A covers the Heston stochastic-volatility model with two contracts, an arithmetic Asian call and a discretely monitored down-and-out call (DOC), each evaluated at full and reduced feature sets. Group B covers geometric Brownian motion with a floating-strike lookback call, again at full and reduced features. The experiments are presented in the order Group B, Group A, beginning with the lookback call because an analytical benchmark is available there that provides an external reference point absent from the Heston experiments.

In every experiment, all three architectures are trained on $N_{\text{train}} = 50,000$ paths with a mini-batch of $N_{\text{batch}} = 10,000$ paths per gradient step, for 3,000 epochs. The reported CVaR and gain statistics are computed on a held-out test pool of $N_{\text{test}} = 50,000$ paths (seed 3), never seen during training. Best-checkpoint restoration is applied: across training, the model state at the epoch with the lowest eval-set CVaR is retained, and final evaluation is performed with these weights. All CVaR values are reported at the $\alpha = 5\%$ tail level. Because CVaR here is defined as the mean of the worst 5% of terminal *gains*, it takes negative values; a less negative value corresponds to better tail performance.

A complementary set of experiments in which the policy’s own previous hedge position is appended to the feature vector is reported in Chapter 7.

6.1 GBM Lookback Call

6.1.1 Full Feature Set (Experiment 5)

The contract is a floating-strike lookback call with payoff $G_T = S_T - \min_k S_{t_k}$ under geometric Brownian motion with $\sigma = 0.20$, $r = 0$, $T = 1$, $n = 50$ hedging steps. The Monte Carlo discrete-path premium is $p_0 = 0.1365$ (100,000 paths, seed 0); the Goldman-Sosin-Gatto continuous-monitoring price is 0.1101, so the discretisation bias amounts to 0.0263. All three neural network models are trained with the MC discrete premium. The Goldman-Sosin-Gatto (GSG) analytical delta is evaluated with the same MC premium. The feature vector is $\mathbf{x}_{t_k} = (S_{t_k}/K, m_{t_k}/K, \tau_k)$, where $m_{t_k} = \min_{j \leq k} S_{t_j}$ is the running minimum explicitly provided as a feature.

Table 8 reports the out-of-sample test results.

Table 8: Test-set results for Experiment 5 (GBM lookback call, full features, $N_{\text{test}} = 50,000$ paths). CVaR is the mean of the worst 5% of terminal gains. Mean cost is the mean total proportional transaction cost. Less negative CVaR is better.

Model	CVaR	Mean gain	Std	Mean cost	Params
Analytical (GSG)	−0.0538	−0.0041	0.0205	0.0041	—
MLP	−0.0390	−0.0028	0.0188	0.0029	4,481
LSTM	−0.0361	−0.0028	0.0181	0.0029	51,009
Transformer	−0.0357	−0.0031	0.0170	0.0031	103,425

All three neural network models outperform the analytical benchmark on CVaR. This is not a contradiction of the GSG optimality result: the GSG formula is optimal for *continuously* monitored contracts under no transaction costs, and applying it to a 50-step

discrete hedging problem incurs a discretisation error that the neural networks avoid by training end-to-end on the discrete payoff.

Among the neural networks, the Transformer achieves the lowest CVaR (-0.0357), followed closely by the LSTM (-0.0361) and then the MLP (-0.0390). The differences are small: LSTM and Transformer are essentially comparable, and the MLP closes most of the gap. The main residual advantage of the Transformer is a lower gain standard deviation (0.0170 versus 0.0188 for the MLP), indicating a tighter hedging distribution.

Figure 1 shows the gain distributions and tail zoom. All three neural network distributions are concentrated and nearly centred, with the Transformer distribution being narrower. The analytical benchmark distribution is wider, consistent with the discretisation error discussed before.

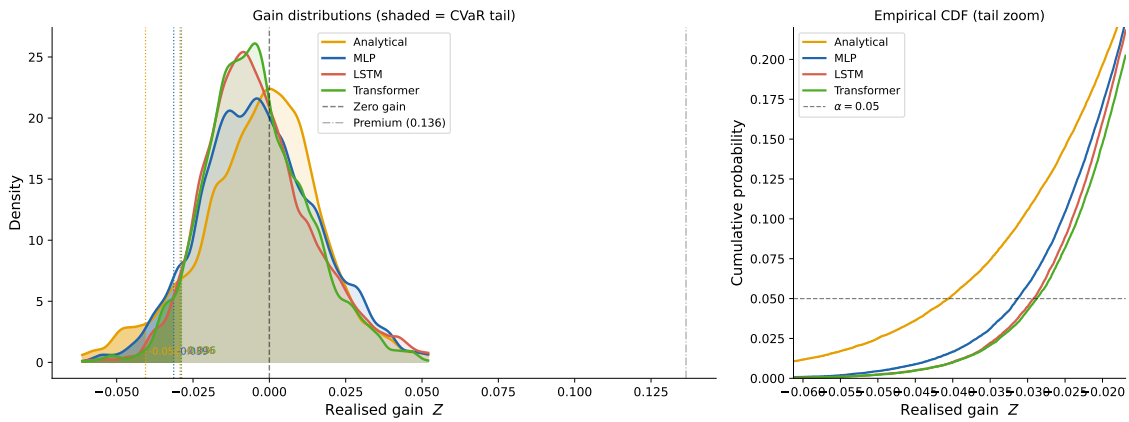


Figure 1: Gain distributions for Experiment 5 (GBM lookback call, full features). Left: kernel density estimates with the CVaR tail shaded; right: empirical CDF tail zoom. Less negative CVaR corresponds to a CDF crossing $\alpha = 0.05$ further to the right.

Table 9 tracks eval CVaR at selected epochs. All three architectures converge smoothly and without instability. The MLP plateaus early: its eval CVaR changes by less than 0.001 between epoch 2,400 and epoch 3,000, while the LSTM and Transformer continue to improve gradually through the full run. Train and eval curves track closely throughout, indicating no overfitting.

6.1.2 Reduced Feature Set (Experiment 6)

The reduced experiment removes the running minimum m_{t_k}/K from the feature vector, leaving only the current price ratio S_{t_k}/K and normalised time-to-maturity τ_k . The MLP is now structurally unable to recover the running minimum: with no memory and no direct feature for m_{t_k} , it can only output a hedge ratio based on the current price and time. The LSTM and Transformer, by contrast, retain access to the full price history through their internal state and attention mechanism respectively.

Table 10 reports the test results.

The results exhibit the largest architecture gap of all six experiments. The MLP CVaR degrades sharply from -0.0390 to -0.0714 , worse than the analytical benchmark (-0.0538) and substantially worse than the temporal models. The MLP training curve confirms the structural failure: the eval CVaR plateaus near 0.072 after the first few hundred epochs and

Table 9: Training progress for Experiment 5 (GBM lookback call, full features). Eval CVaR at selected epochs is the negated CVaR used as the minimisation objective. The best-checkpoint row records the lowest eval CVaR achieved across all epochs; test CVaR is evaluated on the 50,000-path test pool after best-checkpoint restoration (negative; less negative is better).

Epoch	Eval CVaR (negated, ↓)		
	MLP	LSTM	Transformer
200	0.0733	0.0821	0.0649
600	0.0582	0.0572	0.0492
1200	0.0467	0.0439	0.0378
1800	0.0416	0.0389	0.0366
2400	0.0410	0.0378	0.0366
3000	0.0398	0.0374	0.0362
Best checkpoint	0.0398	0.0367	0.0362
Test CVaR	−0.0390	−0.0361	−0.0357

Table 10: Test-set results for Experiment 6 (GBM lookback call, reduced features). The analytical benchmark uses the full three-feature matrix and is reported as a reference only; it does not use the reduced feature set.

Model	CVaR	Mean gain	Std	Mean cost	Params
Analytical (GSG)	−0.0538	−0.0041	0.0205	0.0041	—
MLP	−0.0714	−0.0016	0.0298	0.0017	4,417
LSTM	−0.0411	−0.0028	0.0181	0.0029	50,753
Transformer	−0.0378	−0.0031	0.0177	0.0032	103,361

does not improve meaningfully over the remaining 2,800 epochs, because the information required to hedge optimally simply is not present in the two available features.

The LSTM and Transformer, by contrast, maintain performance close to their full-feature counterparts. The LSTM CVaR moves from −0.0361 (full) to −0.0411 (reduced), a modest deterioration of 0.005. The Transformer CVaR moves from −0.0357 to −0.0378, an even smaller change. Both temporal models therefore recover most of the information contained in m_{t_k} from the price history $(S_{t_0}, \dots, S_{t_k})$.

A notable feature of the reduced experiment is that the LSTM and Transformer outperform the GSG analytical benchmark (−0.0411 and −0.0378 versus −0.0538), despite never observing the running minimum explicitly. This confirms that the temporal models learn a strategy that is better adapted to discrete monitoring than the continuous-time GSG formula, even when the key path statistic is withheld.

Figure 2 shows the gain distributions; the MLP distribution is visibly wider. Table 11 tracks the training progress: the MLP eval CVaR plateaus near 0.072 from the first few hundred epochs onward and barely changes over the remaining 2,800 epochs, reflecting the structural information barrier rather than a training deficiency.

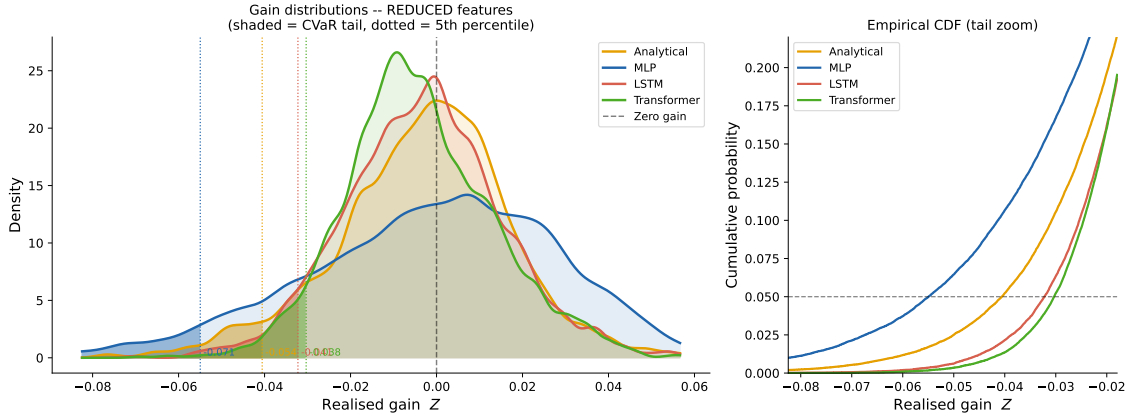


Figure 2: Gain distributions for Experiment 6 (GBM lookback call, reduced features).

Table 11: Training progress for Experiment 6 (GBM lookback call, reduced features). Conventions as in Table 9.

Epoch	Eval CVaR (negated, ↓)		
	MLP	LSTM	Transformer
200	0.0771	0.0881	0.0714
600	0.0754	0.0629	0.0574
1200	0.0729	0.0490	0.0430
1800	0.0722	0.0437	0.0396
2400	0.0721	0.0418	0.0379
3000	0.0722	0.0416	0.0380
Best checkpoint	0.0721	0.0416	0.0378
Test CVaR	−0.0714	−0.0411	−0.0378

6.2 Heston Down-and-Out Call

The down-and-out call (DOC) under the Heston model is hedged with two instruments: the stock S and a variance swap (VS). Parameters are $S_0 = 1.0$, $K = 1.0$, $H = 0.85$, $V_0 = 0.04$, $\kappa = 2.0$, $\theta = 0.04$, $\varepsilon = 0.5$, $\rho = -0.7$, $r = 0$, $T = 1.0$ year, $n = 50$ hedging steps. The MC premium is $p_0 = 0.0676$ (100,000 paths, seed 0). No analytical delta is available; the three neural network architectures are compared exclusively against each other.

6.2.1 Full Feature Set (Experiment 3)

The full feature vector is $\mathbf{x}_{t_k} = (S_{t_k}/S_0, \text{VS}_{t_k}/\text{VS}_0, m_{t_k}/H, S_{t_k}/H, \tau_k)$, where m_{t_k}/H is the running minimum scaled by the barrier, serving as a continuous alive indicator.

Table 12 reports the test results.

The LSTM and Transformer achieve nearly identical CVaR values (-0.02635 and -0.02630 respectively), both substantially better than the MLP (-0.03211). The gap between the temporal models and the MLP (≈ 0.006) is material: it represents a 18% improvement in tail CVaR. The LSTM achieves this with slightly lower mean transaction costs (0.0023 versus 0.0027 for the Transformer), while the Transformer has a marginally worse mean

Table 12: Test-set results for Experiment 3 (Heston DOC, full features).

Model	CVaR	Mean gain	Std	Mean cost	Params
MLP	−0.0321	−0.0032	0.0254	0.0025	4,674
LSTM	−0.0263	−0.0029	0.0214	0.0023	51,586
Transformer	−0.0263	−0.0033	0.0224	0.0027	103,618

gain (−0.0033 versus −0.0029).

Table 13 tracks the training progress. Both temporal models converge to a comparably low CVaR level: the best eval values are 0.02606 for the LSTM and 0.02605 for the Transformer, a difference negligible given the estimation noise from 10,000 evaluation paths. The MLP best checkpoint is 0.03189, substantially higher despite a smooth training curve, confirming that the architecture ceiling is lower rather than that training was insufficient.

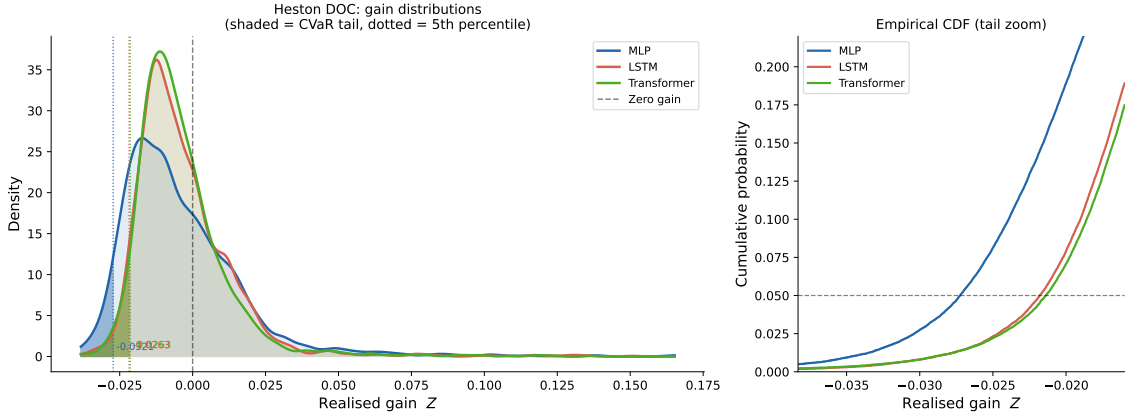


Figure 3: Gain distributions for Experiment 3 (Heston DOC, full features). The LSTM and Transformer distributions are visually indistinguishable; the MLP distribution is wider.

6.2.2 Reduced Feature Set (Experiment 4)

The reduced feature vector removes the running minimum m_{t_k}/H , retaining $(S_{t_k}/S_0, VS_{t_k}/VS_0, S_{t_k}/H, \tau,$

Table 14 reports the test results.

The pattern from the full experiment is almost exactly reproduced. LSTM and Transformer remain nearly identical (−0.02650 and −0.02645), and the MLP worsens only marginally from −0.0321 to −0.0332. The deterioration from removing m_{t_k}/H is small across all three models, suggesting that the running minimum for the barrier option is relatively easy to infer from the price path: a knockout is a permanent and discrete event, and its occurrence can be tracked implicitly from the observed price sequence. This is in contrast to the lookback experiment, where the running minimum is a continuously updated sufficient statistic that the MLP genuinely cannot reconstruct.

Figure 4 shows the gain distributions; the pattern mirrors the full-feature experiment closely. Table 15 tracks training progress. The Transformer starts with a slightly elevated eval CVaR at epoch 200 relative to the MLP but recovers by epoch 600 and converges to the same level as the LSTM.

Table 13: Training progress for Experiment 3 (Heston DOC, full features). Conventions as in Table 9.

Epoch	Eval CVaR (negated, ↓)		
	MLP	LSTM	Transformer
200	0.0554	0.0557	0.0612
600	0.0369	0.0357	0.0354
1200	0.0354	0.0294	0.0283
1800	0.0346	0.0268	0.0269
2400	0.0329	0.0274	0.0265
3000	0.0319	0.0261	0.0278
Best checkpoint	0.0319	0.0261	0.0261
Test CVaR	−0.0321	−0.0263	−0.0263

Table 14: Test-set results for Experiment 4 (Heston DOC, reduced features).

Model	CVaR	Mean gain	Std	Mean cost	Params
MLP	−0.0332	−0.0034	0.0281	0.0027	4,610
LSTM	−0.0265	−0.0029	0.0204	0.0023	51,330
Transformer	−0.0265	−0.0033	0.0224	0.0026	103,554

6.3 Heston Arithmetic Asian Call

The arithmetic Asian call has payoff $G_T = (\bar{S} - K)^+$ where $\bar{S} = n^{-1} \sum_{k=1}^n S_{t_k}$. Parameters are $S_0 = 100$, $K = 100$, $V_0 = 0.04$, $\kappa = 1$, $\theta = 0.04$, $\varepsilon = 2$, $\rho = -0.7$, $r = 0$, $T = 30/365$, $n = 30$ hedging steps. The MC premium is $p_0 = 1.1567$ (200,000 paths, seed 0). Gain and CVaR values are reported in the same units as the underlying stock price. The lookback and barrier experiments use $S_0 = 1$, so the figures in Sections 6.1 and 6.2 read directly as fractions of the initial spot. The Asian experiments use $S_0 = 100$, so the figures in Section 6.3 are larger by a factor of 100 in absolute terms; cross-experiment comparisons require dividing by the corresponding S_0 .

6.3.1 Full Feature Set (Experiment 1)

The full feature vector is $\mathbf{x}_{t_k} = (S_{t_k}/S_0, \text{VS}_{t_k}/\text{VS}_0, A_{t_k}/S_0, \tau_k)$, where $A_{t_k} = k^{-1} \sum_{j=1}^k S_{t_j}$ is the running arithmetic average (with $A_{t_0} = S_{t_0}$ at $k = 0$).

Table 16 reports the test results.

The Asian experiments show the clearest ranking across all six experiments. The Transformer achieves a CVaR of -0.9504 , compared to -1.1834 for the LSTM and -1.4354 for the MLP, a gap of approximately 0.25 between each consecutive pair. This improvement in tail risk comes at a visible cost: the Transformer has the worst mean gain (-0.1361 versus -0.1052 for the MLP) and the highest mean transaction costs (0.1341 versus 0.1022 for the MLP). The Transformer is therefore trading more actively, incurring higher costs in expectation, but achieving a substantially tighter tail through finer control of the hedge ratio. The gain standard deviation confirms this: 0.3932 for the Transformer versus 0.5847 for the MLP.

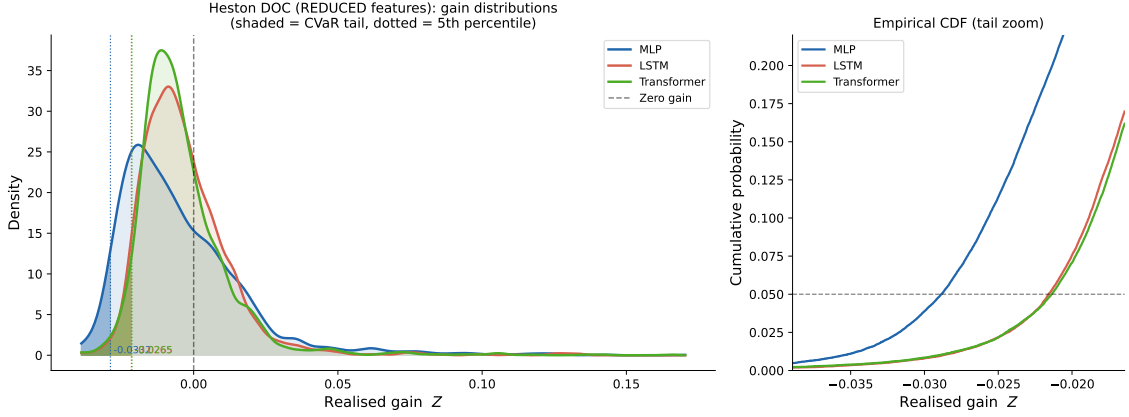


Figure 4: Gain distributions for Experiment 4 (Heston DOC, reduced features). Results are very similar to the full-feature experiment, indicating that removal of the running minimum feature has little practical effect.

Table 15: Training progress for Experiment 4 (Heston DOC, reduced features). Conventions as in Table 9.

Epoch	Eval CVaR (negated, ↓)		
	MLP	LSTM	Transformer
200	0.0461	0.0557	0.0473
600	0.0378	0.0399	0.0408
1200	0.0364	0.0304	0.0291
1800	0.0360	0.0277	0.0274
2400	0.0345	0.0318	0.0264
3000	0.0332	0.0271	0.0259
Best checkpoint	0.0332	0.0265	0.0258
Test CVaR	−0.0332	−0.0265	−0.0265

Figure 5 shows the gain distributions. Table 17 tracks the training progress. All three architectures are still decreasing at epoch 3,000: the Transformer moves from 5.639 at epoch 200 to 0.955 at the best checkpoint, a reduction of more than 80%.

6.3.2 Reduced Feature Set (Experiment 2)

The reduced feature vector removes the running average A_{t_k}/S_0 , leaving $(S_{t_k}/S_0, VS_{t_k}/VS_0, \tau_k)$.

Table 18 reports the test results.

The degradation from removing the running average is modest across all three architectures: CVaR worsens by 0.034 for the MLP, 0.024 for the LSTM, and 0.067 for the Transformer relative to the full-feature results. The relative ranking (Transformer best, MLP worst) is preserved.

The small effect of removing A_{t_k}/S_0 contrasts sharply with the lookback experiment, where removing m_{t_k}/K caused the MLP CVaR to approximately double. A plausible explanation is that the arithmetic running average is less information-dense than the running minimum

Table 16: Test-set results for Experiment 1 (Heston Asian call, full features). Figures are in units of the stock price ($S_0 = 100$).

Model	CVaR	Mean gain	Std	Mean cost	Params
MLP	-1.4354	-0.1052	0.5847	0.1022	4,610
LSTM	-1.1834	-0.1104	0.4840	0.1086	51,330
Transformer	-0.9504	-0.1361	0.3932	0.1341	106,690

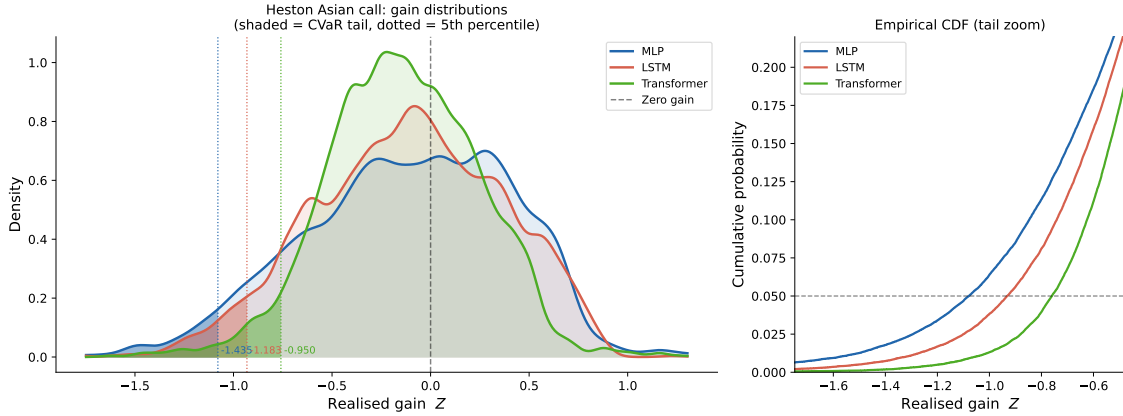


Figure 5: Gain distributions for Experiment 1 (Heston Asian call, full features). The Transformer distribution is noticeably narrower, consistent with the lower standard deviation and improved CVaR.

as a path statistic: the average evolves slowly and is partially predictable from the current price level and time, whereas the minimum can jump discontinuously relative to prior values and is harder to reconstruct from the visible features alone. All three architectures, including the memoryless MLP, are therefore only mildly affected by its omission.

Figure 6 shows the gain distributions; the pattern mirrors the full-feature experiment closely. Table 19 tracks training progress. The LSTM experiences a notable anomaly at epoch 3,000: the training loss spikes from ≈ 1.195 at epoch 2,800 to 2.008, destabilising the weights, and the eval CVaR correspondingly degrades from 1.202 to 1.308 over the same interval. The best-checkpoint mechanism automatically restores the epoch-2,800 weights for the final evaluation, demonstrating the practical importance of tracking the best evaluation checkpoint rather than using the final-epoch weights.

Table 17: Training progress for Experiment 1 (Heston Asian call, full features). Conventions as in Table 9. Figures are in units of the stock price ($S_0 = 100$).

Epoch	Eval CVaR (negated, $\downarrow$)		
	MLP	LSTM	Transformer
200	5.5283	5.3635	5.6391
600	3.2461	3.1217	3.3050
1200	1.9926	1.6968	1.5449
1800	1.6266	1.3661	1.2099
2400	1.5164	1.2521	1.0664
3000	1.4273	1.1784	0.9548
Best checkpoint	1.4273	1.1784	0.9548
Test CVaR	-1.4354	-1.1834	-0.9504

Table 18: Test-set results for Experiment 2 (Heston Asian call, reduced features).

Model	CVaR	Mean gain	Std	Mean cost	Params
MLP	-1.4690	-0.1144	0.6174	0.1121	4,546
LSTM	-1.2069	-0.1053	0.4961	0.1037	51,074
Transformer	-1.0175	-0.1352	0.4010	0.1336	106,626

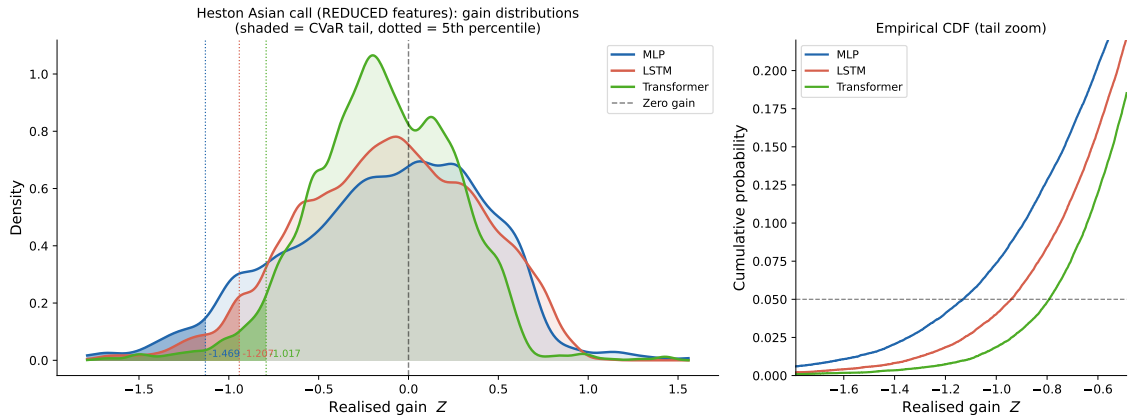


Figure 6: Gain distributions for Experiment 2 (Heston Asian call, reduced features). The pattern mirrors the full-feature experiment closely, consistent with the small degradation reported in Table 18.

Table 19: Training progress for Experiment 2 (Heston Asian call, reduced features). Conventions as in Table 9. Figures are in units of the stock price ($S_0 = 100$). The LSTM epoch-3,000 eval CVaR is elevated due to a late-epoch training instability; the best checkpoint is from epoch 2,800.

Epoch	Eval CVaR (negated, $\downarrow$)		
	MLP	LSTM	Transformer
200	5.5387	5.4986	5.5427
600	3.3467	3.1206	4.1469
1200	2.1016	1.9370	1.6767
1800	1.6503	1.3573	1.1694
2400	1.5292	1.2639	1.0787
3000	1.4689	1.3084 [†]	1.0172
Best checkpoint	1.4689	1.2025	1.0172
Test CVaR	-1.4690	-1.2069	-1.0175

[†] LSTM eval CVaR increased from 1.202 (epoch 2,800) to 1.308 (epoch 3,000) due to a training-loss spike; best-checkpoint restoration uses epoch-2,800 weights.

7 Position-Aware Hedging Policies

The numerical experiments of Chapter 6 treat the hedging policy as a function of the observable market state alone. Under proportional transaction costs, however, the optimal Markovian policy depends also on the policy’s own previous position.

This chapter reports a controlled ablation in which the previous hedge ratio is appended to the input feature vector across all six experiments, with every other element of the setup, hyperparameters, seeds, network architectures, training procedure, and Monte Carlo paths, held identical to Chapter 6. The aim is to test whether the theoretical sufficient statistic translates into an empirical improvement in tail-risk quality at the transaction-cost coefficient used throughout ($c = 0.001$).

7.1 Setup

For the two-instrument Heston experiments the augmented feature vector at step k is

$$\mathbf{x}_{t_k}^{\text{PD}} = (\mathbf{x}_{t_k}, \delta_{t_{k-1}}^S, \delta_{t_{k-1}}^{\text{VS}}), \quad (7.1)$$

where $\mathbf{x}_{t_k}$ is the original feature vector for the experiment in question, as defined in Chapter 6, and $\delta_{t_{k-1}}^S, \delta_{t_{k-1}}^{\text{VS}}$ are the policy’s stock and variance-swap positions at the previous step (both initialised to zero at $k = 0$). For the single-instrument lookback experiments only the scalar $\delta_{t_{k-1}}$ is appended.

Because $\delta_{t_{k-1}}$ is itself the output of the policy at the previous step, the augmented feature matrix cannot be precomputed and the loss becomes intrinsically sequential. At each step k the policy network is evaluated with the exact previous-step output in its input, without any gradient detachment, so the gradient of the terminal CVaR loss propagates through the entire T -step chain $\delta_{t_{T-1}} \rightarrow \delta_{t_{T-2}} \rightarrow \dots \rightarrow \delta_{t_0} \rightarrow \theta$, giving a backpropagation-through-time problem of depth $T \in \{30, 50\}$.

All experimental hyperparameters are identical to those of Chapter 6: $N_{\text{train}} = 50,000$, $N_{\text{test}} = 50,000$, $N_{\text{batch}} = 10,000$, 3,000 epochs, learning rate 10^{-3} , AdamW with weight decay 10^{-4} , $\alpha = 0.05$, $c = 0.001$, and unchanged Monte Carlo seeds. The networks retain the architectures and hidden dimensions of Chapter 5; only the width of the input layer changes to accommodate the additional features.

7.2 Aggregate Results

Table 20 reports the test-set CVaR for every experiment and architecture, both with and without the previous-position feature. The column Δ records the change relative to the baseline: $\Delta < 0$ indicates a worse (more negative) CVaR under the prev-delta variant.

The two Asian experiments display the largest and most consistent effect: every architecture degrades by between 0.32 and 0.62 in CVaR. The degradation is largest for the Transformer, which loses essentially all of the baseline advantage it had over the MLP and LSTM. In the barrier and lookback experiments, the differences are at the level of the third or fourth decimal place across all architectures, well within the estimation noise of a 50,000-path test pool, with no consistent sign across architectures or between full and reduced variants.

Table 20: Test-set CVaR with and without the previous hedge position as an additional input feature, all six experiments and three architectures. Less negative CVaR is better. $\Delta = \text{CVaR}^{\text{PD}} - \text{CVaR}^{\text{baseline}}$: $\Delta < 0$ denotes a degradation under prev-delta. Values are in the same units as the corresponding experiment in Chapter 6.

Model	Baseline CVaR	Prev-delta CVaR	Δ
1. Heston Asian, full features			
MLP	-1.4354	-1.7608	-0.3254
LSTM	-1.1834	-1.5705	-0.3871
Transformer	-0.9504	-1.5712	-0.6208
2. Heston Asian, reduced features			
MLP	-1.4690	-1.8059	-0.3369
LSTM	-1.2069	-1.5889	-0.3820
Transformer	-1.0175	-1.5772	-0.5597
3. Heston DOC, full features			
MLP	-0.0321	-0.0324	-0.0003
LSTM	-0.0263	-0.0255	+0.0008
Transformer	-0.0263	-0.0252	+0.0011
4. Heston DOC, reduced features			
MLP	-0.0332	-0.0326	+0.0006
LSTM	-0.0265	-0.0262	+0.0003
Transformer	-0.0265	-0.0259	+0.0006
5. GBM lookahead, full features			
MLP	-0.0390	-0.0372	+0.0018
LSTM	-0.0361	-0.0363	-0.0002
Transformer	-0.0357	-0.0364	-0.0007
6. GBM lookahead, reduced features			
MLP	-0.0714	-0.0721	-0.0007
LSTM	-0.0411	-0.0402	+0.0009
Transformer	-0.0378	-0.0390	-0.0012

7.3 The Asian Case in Detail

Two patterns stand out in the training dynamics of the full-feature Asian experiment under prev-delta. First, all three prev-delta variants converge much more slowly than their baseline counterparts: at epoch 3,000 the LSTM eval CVaR sits near 1.56 in the prev-delta case, compared to 1.18 in the baseline (Table 17). Second, the three architecture curves cluster together under prev-delta, in contrast to the baseline where the Transformer separates clearly from the LSTM and MLP by the midpoint of training. The Transformer’s structural advantage in the Asian case is suppressed when its own previous output is fed back as an input.

Figure 7 shows the corresponding gain distributions on the test pool. The tails are visibly heavier than in the baseline (Figure 5), and the distributions of the three architectures are much closer to each other.

A useful diagnostic is the mean transaction cost incurred by each policy. Under prev-delta the mean cost falls from 0.1022 to 0.0758 for the MLP, from 0.1086 to 0.0854 for the LSTM, and from 0.1341 to 0.1189 for the Transformer. Each policy trades less actively when given access to its previous position, but the reduction is not compensated by improvements

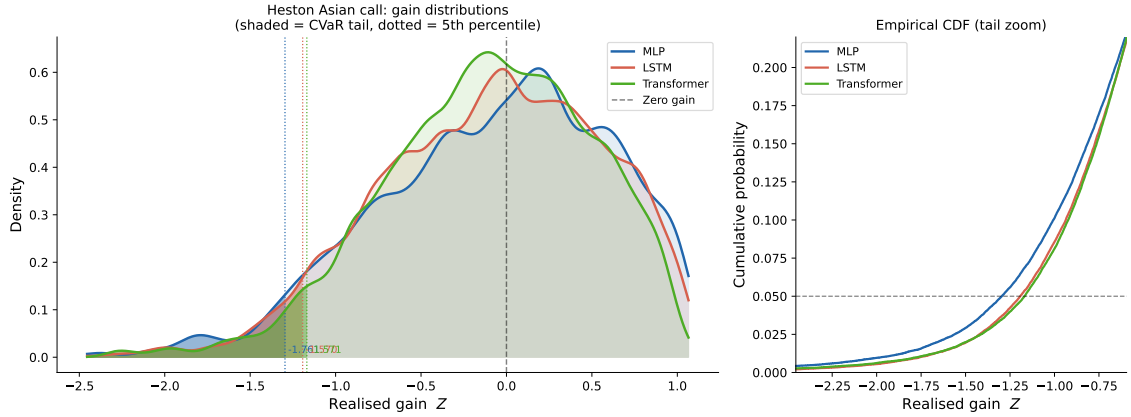


Figure 7: Gain distributions for the prev-delta variant of Experiment 1. Compared to the baseline (Figure 5), the tails are heavier and the distributions less well-separated by architecture.

elsewhere: the gain standard deviation rises in parallel and the net effect on the tail is uniformly negative. The prev-delta networks settle on policies that are more conservative about rebalancing, paying lower costs but accepting wider gain distributions, which is the opposite of the trade-off the CVaR objective should encourage. A fuller interpretation of these patterns is given in Section 8.4.

7.4 The Barrier and Lookback Cases

In the barrier and lookback experiments the prev-delta differences are within estimation noise. Figure 8 shows the gain distributions for the full-feature barrier experiment under prev-delta; they are visually indistinguishable from those of the baseline.

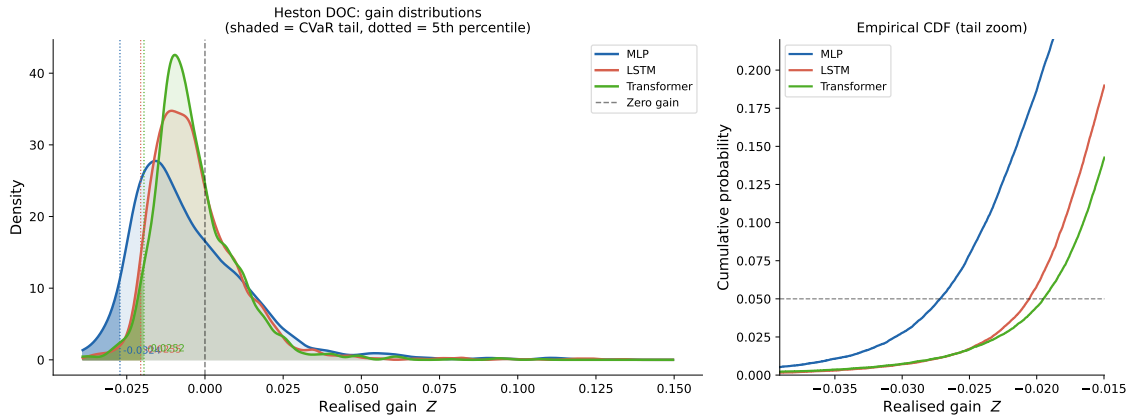


Figure 8: Gain distributions for the prev-delta variant of Experiment 3 (Heston DOC, full features). Visually indistinguishable from the baseline.

Both contracts incur much lower absolute transaction costs than the Asian: roughly 0.002 to 0.003 per path for the barrier and lookback, against roughly 0.10 to 0.13 for the Asian. Both payoffs also depend on the path through a simple statistic, the barrier-crossing flag or the running minimum, which the temporal architectures already extract from the price history in the baseline experiments.

8 Discussion and Conclusion

This chapter synthesises the results of the six experiments of Chapter 6 and the position-aware ablation of Chapter 7, addresses the central question of whether temporal memory in the neural network architecture provides a meaningful hedging advantage, and discusses why providing the policy with its own previous hedge position fails to improve tail-risk quality at the transaction-cost coefficient studied. The chapter closes with a discussion of limitations and directions for future work.

8.1 Does Temporal Memory Help?

Across all six experiments the answer is consistently yes, but the magnitude varies substantially with contract type and feature regime.

The reduced-feature experiments make the role of memory clearest. In the GBM lookback (Experiment 6), the MLP eval CVaR plateaus near 0.072 within a few hundred epochs and does not improve thereafter: with only the current price and time as inputs, no further training can recover the running minimum that drives the payoff. The LSTM and Transformer reach -0.0411 and -0.0378 on test, both better than the GSG analytical benchmark (-0.0538). The Asian (Experiment 2) shows a milder gradient: removing A_{t_k}/S_0 widens CVaR by 0.034 for the MLP, 0.024 for the LSTM, and 0.067 for the Transformer; the running average evolves slowly and is partially predictable from the current price and elapsed time, so its omission hurts less. The barrier (Experiment 4) is the most forgiving: removing m_{t_k}/H moves the MLP CVaR only from -0.0321 to -0.0332 , with comparable shifts for the temporal models, because a knockout is a permanent, discrete event that the current price level relative to the barrier S_{t_k}/H already encodes.

The full-feature experiments show that the benefit of memory is not only the ability to recover withheld statistics. With the full feature set the temporal models still beat the MLP in the barrier (-0.0263 versus -0.0321) and in the Asian (-0.9504 Transformer, -1.1834 LSTM, -1.4354 MLP), even though all three architectures see the same path-dependent state. Attending to the full price history when deciding how to rebalance is itself useful, independently of any sufficient-statistic argument.

8.2 Transformer versus LSTM

The Transformer either matches or outperforms the LSTM across all six experiments, but the margin varies by contract type.

In the Asian call experiments, the Transformer is clearly superior to the LSTM: in Experiment 1, it achieves a CVaR of -0.9504 against -1.1834 , a difference of 0.23 in absolute terms. This is the largest architecture gap in the study. The Asian payoff depends on the entire path through the arithmetic average, and there is no simple sufficient statistic of fixed dimension that summarises the relevant history. Self-attention, which can in principle attend directly to any past step with a learned weighting, is better suited to aggregating this diffuse historical dependence than the LSTM's fixed-size hidden state, which must compress arbitrarily long path histories into a 64-dimensional vector.

In the barrier experiments, the Transformer and LSTM are essentially equivalent. The best eval CVaR values in Experiment 3 are 0.02605 (Transformer) and 0.02606 (LSTM), a difference well within the estimation noise from 10,000 evaluation paths. The DOC payoff depends on the path only through the running minimum, which is a low-dimensional

sufficient statistic that an LSTM can retain accurately in its cell state. There is no diffuse historical dependence for the attention mechanism to exploit, so the two architectures perform equivalently.

In the lookback experiments, the Transformer holds a small consistent advantage: the CVaR gap between Transformer and LSTM is 0.0003 in Experiment 5 and 0.0032 in Experiment 6. The lookback payoff depends on the path through the running minimum, but the minimum is updated continuously and has a finer structure than the discrete knockout event. Self-attention can exploit this structure by directly inspecting the lowest observed price at any past step, whereas the LSTM must track the minimum implicitly through its hidden state.

Overall, the Transformer is at least as capable as the LSTM in every setting and provides a meaningful improvement wherever the payoff depends on diffusely distributed path information.

8.3 What the Gain Decomposition Reveals

The gain decomposition separates terminal gains into the hedging profit-and-loss component and the transaction cost component. For both the Asian and lookback experiments, the Transformer incurs higher mean transaction costs than the MLP. In Experiment 1, mean costs are 0.1341 (Transformer) versus 0.1022 (MLP); in Experiment 5, 0.0031 versus 0.0029. The Transformer trades more actively, rebalancing the hedge ratio more frequently or by larger amounts. The benefit is a substantially tighter gain distribution: the gain standard deviation in Experiment 1 is 0.3932 (Transformer) against 0.5847 (MLP), corresponding to the large improvement in CVaR. The Transformer is willing to absorb higher expected costs in exchange for a better tail outcome, which is exactly the trade-off that CVaR optimisation is designed to encourage.

In the barrier experiments, the LSTM consistently achieves slightly lower mean costs than the Transformer (0.0023 versus 0.0027 in Experiment 3), while reaching essentially the same CVaR level. This suggests that the LSTM and Transformer find slightly different policies with similar tail quality but different average cost profiles.

8.4 Augmenting the Information Set with the Previous Hedge Position

The position-aware ablation of Chapter 7 appends the previous hedge ratio to the feature vector across all six experiments. At $c = 0.001$ it does not help. In the Asian experiments it degrades CVaR by between 0.32 and 0.62 across architectures, with the Transformer hit hardest and losing essentially all of its baseline advantage over the simpler models. In the barrier and lookback experiments the effect is within estimation noise.

Three reinforcing factors explain the Asian degradation. First, the sequential exact loss imposes a backpropagation chain of length $T = 30$ on every gradient step, amplifying gradient variance and slowing convergence relative to the single-pass forward used in the baseline; the 3,000-epoch budget, already insufficient for the Asian baseline, is more clearly insufficient here. Second, the do-not-trade policy $\delta_{t_k} = \delta_{t_{k-1}}$ is a local minimum of the transaction-cost term, and providing the previous position as an explicit input makes this near-identity map easy to parametrise; mean transaction costs in the Asian drop materially under prev-delta (from 0.1022 to 0.0758 for the MLP and from 0.1341 to 0.1189 for the Transformer), consistent with the networks partially settling on the attractor. Third, at $c = 0.001$ the frictionless delta, which depends on the current market state alone, is close to

the cost-aware optimal hedge: the marginal informational content of the previous position is small and is outweighed by the deeper gradient chain it introduces.

The neutrality on the barrier and lookback is consistent with the same logic at a different cost regime. Both contracts spend a smaller share of their premium on transaction costs (roughly 4% for the barrier and 2% for the lookback, against roughly 10% for the Asian), so the do-not-trade attractor is weaker and the extra optimisation burden no longer dominates. Both also have simpler path dependence than the temporal architectures already extract from the baseline; the previous position adds little.

The Transformer takes the largest absolute hit because its baseline advantage came from attending cleanly across the price history; under prev-delta that history is contaminated by the policy’s own past outputs, and the backpropagation-through-time problem inherits the depth of the entire hedging horizon.

8.5 Limitations

Several limitations of the current study should be noted.

The hedging horizon is short: 50 steps for the lookback and barrier experiments and 30 steps for the Asian experiments. In practice equity options are often hedged daily over horizons of several months, with time-series lengths an order of magnitude larger. The quadratic attention complexity $O(T^2)$ of the Transformer may pose a computational bottleneck at long horizons, whereas the LSTM processes each step in amortised constant time.

The experiments use a fixed set of architecture hyperparameters, with hidden dimension 64, two layers or blocks, and four attention heads across all contracts and architectures. No systematic hyperparameter search was performed; it is possible that a shallower or narrower Transformer performs similarly at lower parameter count, or that a wider LSTM would close the gap in the Asian experiments.

All experiments assume a known parametric model. The networks are trained on simulated paths from the same model used to generate the test set. A more demanding evaluation would train under one model and test under a misspecified alternative, assessing robustness to model error.

The transaction cost coefficient is fixed at $c = 0.001$ across all experiments. Higher transaction costs would penalise the Transformer’s more active rebalancing more severely and might alter the ranking between architectures; they would also widen the no-trade region around the frictionless delta and potentially reverse the verdict of the position-aware ablation.

The barrier and lookback contracts are both monitored only at the discrete trading dates. For the down-and-out call, knockout is checked only against the prices observed at $t_0, t_1, \dots, t_n$; an intra-step path that briefly dips below the barrier and recovers is not recorded as a breach, whereas a continuously monitored DOC would knock out in the same scenario. For the lookback call, the running minimum m_{t_k} is computed over the same discrete grid, and a continuously monitored minimum would be uniformly lower. The CVaR figures reported here therefore correspond to a comparatively benign monitoring schedule; tighter monitoring would raise the hedging difficulty in both contracts and could alter the architecture comparison, especially for the barrier where most of the hedging value sits in correctly anticipating the absorbing event.

8.6 Conclusion

This paper evaluates a position-wise MLP, an LSTM, and a causal Transformer as hedging policies for path-dependent option portfolios optimised under CVaR. The main finding is that temporal memory provides a consistent and sometimes substantial improvement over the memoryless baseline. The Transformer is at least as capable as the LSTM in every setting studied, with a clear advantage wherever the payoff depends on diffusely distributed path information, as in the Asian call experiments.

The reduced-feature experiments provide the sharpest evidence for the role of memory: when the key path statistic is withheld, only memory-capable architectures can recover it from the observed price history, and the MLP training curve flatlines completely while the LSTM and Transformer continue to improve. The full-feature experiments establish that the benefit of memory extends beyond the recovery of withheld statistics: even when all relevant path variables are observed, attending to the full price history improves hedging quality.

The position-aware ablation of Chapter 7 adds an important caveat. Appending the previous hedge position to the feature vector fails to improve hedging quality at $c = 0.001$ and degrades the Asian results substantially. The empirical penalty of the deeper sequential exact loss exceeds the theoretical informational gain at this cost level. Whether the picture reverses at higher transaction costs, where the no-trade region matters more, is an open question and a natural next step for further experimentation.

Taken together, these results suggest that sequence models, and in particular causal Transformers, are a natural architecture for deep hedging problems involving path-dependent payoffs, but that theoretical sufficiency claims about the policy’s information set do not automatically translate into empirical improvements; the optimisation difficulty introduced by feedback through the policy’s own past output can dominate the informational gain. The main directions for future work are extending the experiments to longer horizons, testing robustness under model misspecification, and revisiting the position-aware ablation at higher transaction-cost levels.

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